

Part A: Negative Numbers and Order of Operations*Evaluate each expression completely. Show each major order-of-operations step.*

1. $-10 + 8$

6. $-7 + 8 \cdot (-8)$

2. $2 - (-1)$

7. $-20 + (-2) \cdot (-7)$

3. $-2 \cdot 9$

8. $(-3 + 20)^2 \div 1$

4. $63 \div (-9)$

9. $(-5)^3 - 9 \cdot 5$

5. $-8 + 8 \cdot 9$

10. $3^3 - 4 \cdot 5$

Part B: Fractions*Evaluate. Write every answer in lowest terms.*

1. $1 - \frac{1}{6}$

6. $\frac{5}{9} \div \frac{2}{3}$

2. $\frac{11}{12} - \frac{3}{5}$

7. $2\frac{1}{4} + 3\frac{2}{3}$

3. $2\frac{1}{3} + 3$

8. $1\frac{2}{5} - 1\frac{8}{9}$

4. $\frac{3}{4} \cdot 1\frac{1}{2}$

9. $\frac{4}{10}$

5. $\frac{2}{3} \div 1\frac{1}{10}$

10. $\frac{6}{15}$

Part C: Simplifying and Factoring Expressions*Simplify or factor as directed. Show distribution and like-term steps.*

1. $x - 7 + 3x + 3$

6. $10(6x + 2) + 7x - 10$

2. $5x + 7 + 8x + 6$

7. $10x - 6$

3. $6x^2 + 4x + 5x + 5x^2$

8. $-6x - 4$

4. $6x + 9x^2 + x - 3x^2$

9. $\text{GCF}(7x^3, 56x^2)$

5. $-(-2x + 12) - 6x - 5$

10. $\text{GCF}(18x^3, 9x^4)$

Part D: Solving Linear Equations

Solve each equation for x . Show your steps and check when practical.

1. $8x = -16$

7. $2(4x + 9) + 11 = 53$

2. $2x + 3 = -15$

8. $-x + 9 = 8x + 72$

3. $9x + 9 = 90$

9. $-12x + 5 = -10x - 3$

4. $8x - 2 = -66$

10. $-10x - 7 = 3x - 59$

5. $6(4x + 6) + 9 = -27$

11. $3x + 9 = -8x - 101$

6. $8(8x + 11) - 11 = 141$

12. $\frac{8}{5}x - 20 = \frac{12}{5}$

13. $-\frac{7}{5}x - 1 = -\frac{138}{5}$

14. $\frac{2x - 14}{3} = \frac{-30}{9}$

Part E: Ratios, Proportions, Percents, and Rates

Solve each problem. Show your setup and simplify your answer.

1. $\frac{2}{11} = \frac{x}{33}$

6. 30% of 440

2. $\frac{8}{x} = \frac{40}{75}$

7. from 60 to 57

3. $\frac{4}{2} = \frac{x}{10}$

8. from 10 to 2

4. 20% of 110

9. Noah pays \$90 for 5 phone cases. Find the price per phone case.

5. 60% of 800

10. Option A: \$18 for 6 erasers. Option B: \$24 for 12 erasers. Which option is the better price per eraser?

Part F: Exponent Rules

Simplify completely. Write every power with a positive exponent.

1. $x^5 \cdot x^3$

5. $9 \cdot (6^2)^{-3}$

2. $3x^5 \cdot x^3$

6. $8 \cdot 3^{-4}$

3. $\frac{4x^8}{x^4}$

7. $7^3 \cdot 7^{-1}$

4. $(2x^5)^3$

8. $\left(\frac{2x^4}{x^3}\right)^3$

Part G: Roots and Scientific Notation

Simplify or evaluate completely.

1. $\sqrt{36}$

5. 0.00058

2. $\sqrt[3]{125}$

6. 633

3. $\sqrt{88}$

7. $(1.955 \times 10^4) \div (8.5 \times 10^4)$

4. $\sqrt{40}$

Part H: Linear Inequalities

Solve each inequality for x . Reverse the inequality when required.

1. $4x < 8$

6. $-6x - 6 > 7x - 19$

2. $2x < 8$

7. $12x - 11 \leq -7x + 160$

3. $8x + 8 > 56$

8. $-\frac{7}{3}x + 11 > -24$

4. $9x - 9 > 36$

9. $-\frac{1}{3}x + 13 \geq 7$

5. $2x - 2 \leq 4x + 18$

10. $111 \leq -12x - 21 < 231$

Part I: Slope, Lines, and the Coordinate Plane

Find the requested slope, line equation, or measure.

1. $y = 4x + 8$

3. $(-1, -5)$ and $(6, 58)$

2. $y = 6x - 30$

4. $(-8, -55)$ and $(1, 8)$

9. $m = 3$ through $(1, 4)$

5. $(-8, -15)$ and $(-2, -6)$

10. $m = \frac{1}{2}$ through $(-6, -3)$

6. $(-9, -26)$ and $(6, 19)$

11. $(10, 14)$ and $(18, 20)$

7. $(-5, 1)$ and $(4, 10)$

12. $(-9, 4)$ and $(-19, 6)$

8. $(-7, -69)$ and $(2, 12)$

Part J: Word Problems and Modeling

Define a variable, write a model, solve, and interpret the answer.

1. Zoe buys 6 tickets each week and already owns 22 tickets. After how many weeks will Zoe have 52 tickets?
2. Ines buys 6 candles each week and already owns 5 candles. After how many weeks will Ines have 17 candles?
3. Ethan buys 7 books each week and already owns 24 books. After how many weeks will Ethan have 52 books?
4. Nina rides at a steady 19 miles per hour and travels 38 miles. How many hours does the ride take?
5. Mia rides at a steady 9 miles per hour and travels 36 miles. How many hours does the ride take?

6. A marble originally costs \$120. Ines uses a 15% off coupon. How many dollars does Ines save?
7. A sticker originally costs \$80. Iris uses a 30% off coupon. How many dollars does Iris save?
8. Plan A charges \$20 plus \$4 per month. Plan B charges \$5 plus \$9 per month. After how many months do the two plans cost the same?
9. Plan A charges \$23 plus \$12 per month. Plan B charges \$5 plus \$14 per month. After how many months do the two plans cost the same?

Answer Key

Part A: Negative Numbers and Order of Operations

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|----------|----------|-----------|
| 1. -2 | 5. 64 | 9. -170 |
| 2. 3 | 6. -71 | 10. 7 |
| 3. -18 | 7. -6 | |
| 4. -7 | 8. 289 | |

Part B: Fractions

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|--------------------|---------------------|--------------------|
| 1. $\frac{5}{6}$ | 5. $\frac{20}{33}$ | 9. $\frac{2}{5}$ |
| 2. $\frac{19}{60}$ | 6. $\frac{5}{6}$ | 10. $\frac{32}{5}$ |
| 3. $5\frac{1}{3}$ | 7. $5\frac{11}{12}$ | |
| 4. $1\frac{1}{8}$ | 8. $-\frac{22}{45}$ | |

Part C: Simplifying and Factoring Expressions

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|-----------------|-----------------|------------|
| 1. $4x - 4$ | 5. $-4x - 17$ | 9. $7x^2$ |
| 2. $13x + 13$ | 6. $67x + 10$ | 10. $9x^3$ |
| 3. $11x^2 + 9x$ | 7. $2(5x - 3)$ | |
| 4. $6x^2 + 7x$ | 8. $2(-3x - 2)$ | |

Part D: Solving Linear Equations

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|-------------|-------------|---------------|
| 1. $x = -2$ | 6. $x = 1$ | 11. $x = -10$ |
| 2. $x = -9$ | 7. $x = 3$ | 12. $x = 14$ |
| 3. $x = 9$ | 8. $x = -7$ | 13. $x = 19$ |
| 4. $x = -8$ | 9. $x = 4$ | 14. $x = 2$ |
| 5. $x = -3$ | 10. $x = 4$ | |

Part E: Ratios, Proportions, Percents, and Rates

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|-------------|------------|------------------------|
| 1. $x = 6$ | 5. 480 | 9. \$18 per phone case |
| 2. $x = 15$ | 6. 132 | 10. Option B |
| 3. $x = 20$ | 7. -5% | |
| 4. 22 | 8. -80% | |

Part F: Exponent Rules

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|-----------|---------------------|-----------|
| 1. x^8 | 4. $8x^{15}$ | 7. 49 |
| 2. $3x^8$ | 5. $\frac{1}{5184}$ | 8. $8x^3$ |
| 3. $4x^4$ | 6. $\frac{8}{81}$ | |

Part G: Roots and Scientific Notation

1. 6

2. 5

3. $2\sqrt{22}$

4. $2\sqrt{10}$

5. 5.8×10^{-4}

6. 6.33×10^2

7. 2.3×10^{-1}

Part H: Linear Inequalities

1. $x < 2$

2. $x < 4$

3. $x > 6$

4. $x > 5$

5. $x \geq -10$

6. $x < 1$

7. $x \leq 9$

8. $x < 15$

9. $x \leq 18$

10. $-21 < x \leq -11$

Part I: Slope, Lines, and the Coordinate Plane

1. $m = 4$, $b = 8$, 3. $m = 9$

x -intercept = $(-2, 0)$, 4. $m = 7$

y -intercept = $(0, 8)$, 5. $m = \frac{3}{2}$

2. $m = 6$, $b = -30$, 6. $y = 3x + 1$

x -intercept = $(5, 0)$, 7. $y = x + 6$

y -intercept = $(0, -30)$, 8. $y = 9x - 6$

9. $y = 3x + 1$

10. $y = \frac{1}{2}x$

11. 10

12. $(-14, 5)$

Part J: Word Problems and Modeling

1. 5 weeks

2. 2 weeks

3. 4 weeks

4. 2 hours

5. 4 hours

6. 18 dollars

7. 24 dollars

8. 3 months

9. 9 months